

Final Project Report Template

1. Introduction

1.1. Project overviews - The project “India’s Agricultural Crop Production Analysis Using Tableau” focuses on analyzing agricultural production data of India. It studies crop production, cultivated area, agricultural yield, states, years and seasons. Tableau is used to convert the data into interactive charts, maps and dashboards. The project helps in understanding agricultural trends and comparing production across different categories.

1.2. Objectives - The main objective of this project is to analyze India’s agricultural crop production data and identify important trends and patterns. It aims to compare production across different states, crops, years and seasons. Another objective is to present the findings through clear and interactive Tableau visualizations. The project also helps users understand agricultural performance more easily.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement -

Problem Statement (PS)	I am (student)	I'm trying to	But	Because	Which makes me feel
PS-1	I am a student	I want to analyze India's crop production and identify important trends.	It is difficult to understand and large agricultural data and find useful patterns	I want to make agricultural data simple and useful through interactive dashboards	I like working with data, creating charts, and finding useful insights.
PS-2					

2.2. Project Proposal (Proposed Solution) -

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	The objective is to analyze India's agricultural crop production data. It aims to identify crop-wise and region-wise production trends. The project will present useful insights through Tableau visualizations. It will help users understand agricultural production patterns.
Scope	The project covers agricultural crop production data of India. It includes analysis based on crops, regions and production trends. The data will be processed and visualized using Tableau. The final output will be an interactive and easy-to-understand dashboard.
Problem Statement	
Description	Agricultural data is large and difficult to understand in raw form. It is challenging to identify important production trends manually. Crop-wise and regional differences may not be easily visible. A simple visual analysis is required to understand the data better.
Impact	Difficulty in understanding data can affect decision-making. Farmers and other users may find it difficult to identify useful trends. <u>Government</u> and businesses may also face challenges in analyzing production patterns.

2.3. Initial Project Planning -

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	The objective is to analyze India's agricultural crop production data. It aims to identify crop-wise and region-wise production trends. The project will present useful insights through Tableau visualizations. It will help users understand agricultural production patterns.
Scope	The project covers agricultural crop production data of India. It includes analysis based on crops, regions and production trends. The data will be processed and visualized using Tableau. The final output will be an interactive and easy-to-understand dashboard.
Problem Statement	
Description	Agricultural data is large and difficult to understand in raw form. It is challenging to identify important production trends manually. Crop-wise and regional differences may not be easily visible. A simple visual analysis is required to understand the data better.
Impact	Difficulty in understanding data can affect decision-making. Farmers and other users may find it difficult to identify useful trends. <u>Government</u> and businesses may also face challenges in analyzing production patterns.

					
			Proposed Solution		
			Approach	The agricultural dataset will first be prepared and analyzed. Relevant fields will be selected and visualized using Tableau. Charts and interactive dashboards will be created to show important trends. The final dashboard will provide clear and meaningful insights.	
			Key Features	The project will include interactive Tableau dashboards. It will show crop-wise and region-wise production analysis. Users can explore production trends through charts and filters. The dashboard will present important insights in a simple and attractive format.	

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified -

Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

Data Collection Plan Template

Section	Description
Project Overview	The project focuses on analyzing India's agricultural crop production data. The main objective is to understand crop-wise and region-wise production trends. Tableau will be used to analyze and visualize the data. The final dashboard will present the findings in a simple and interactive way

Data Collection Plan	Agricultural crop production data will be collected for analysis. The required information includes crop, year, state/region and production values. The collected data will be checked for completeness and accuracy. The data will then be prepared for further processing and visualization.
Raw Data Sources Identified	The raw dataset will contain information related to agricultural crop production in India. Important fields such as crop, location, year and production will be identified. The source data will be reviewed before starting the cleaning process. The identified dataset will be used as the base for the Tableau analysis.

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Description of the data in this source.	https://docs.google.com/spreadsheets/d/15nt_b0w_E22-TbmTbieYnL_LqRyyJiiC/edit?usp=drive	excel	15.4 mb	Public 



		_link&ouid=107419396520714809735&rtpof=true&sd=true			
--	--	---	--	--	--

3.2. Data Quality Report -

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset	The dataset may contain missing, duplicate, inconsistent or incorrect values. Some fields may also have incorrect data formats or data types	High	Missing and duplicate values will be identified and corrected or removed. Data formats and values will be standardized to ensure accurate analysis and visualization in Tableau.
---	---	---	---

3.3. Data Exploration and Preprocessing -

Data Exploration and Preprocessing Template

Section	Description
Data Overview	the dataset contains information about India's agricultural crop production. It includes crop, state/region, year and production-related data.
Data Cleaning	Missing, duplicate and incorrect values were identified and removed. The data was checked to improve its accuracy and consistency.
Data Transformation	The data was organized into a suitable format for analysis. Required values and fields were transformed to make visualization easier.
Data Type Conversion	Data types were checked and converted where required. Dates, numerical values and categorical fields were prepared correctly for Tableau.
Column Splitting and Merging	Required columns were split to separate useful information. Related columns were merged where necessary for better analysis.
Data Modeling	The processed data was structured for effective analysis in Tableau.

	Relevant fields were connected and organized to support the dashboard
Save Processed Data	After cleaning and transformation, the processed dataset was saved. The final data was then used for creating Tableau visualizations.

4. Data Visualization

4.1. Framing Business Questions - **Business Questions and Visualisation**

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. What is the trend of agricultural yield from 1997 to 2021?

Description:

This question helps identify how agricultural yield has changed over the years and highlights periods of growth or decline.

2. Which states have the highest agricultural production?

Description:

This helps identify the leading states in agricultural production and compare their overall performance.

3. How is agricultural production distributed across Indian states?

Description:

This question shows the regional distribution of agricultural production across India using a map.

4. Which crops occupy the largest cultivated area?

Description:

This helps compare cultivated areas of different crops and identify crops with higher cultivation coverage.

5. Which states have the largest cultivated area?

Description:

This question compares cultivated areas across states and identifies states with greater agricultural land utilization.

6. How does total agricultural production vary by year?

Description:

This helps understand yearly changes in total agricultural production between 1997 and 2021.

7. How is agricultural production distributed by unit type?

Description:

This shows the distribution of production according to different measurement or unit types in the dataset.

8. Which season contributes the most to agricultural production?

Description:

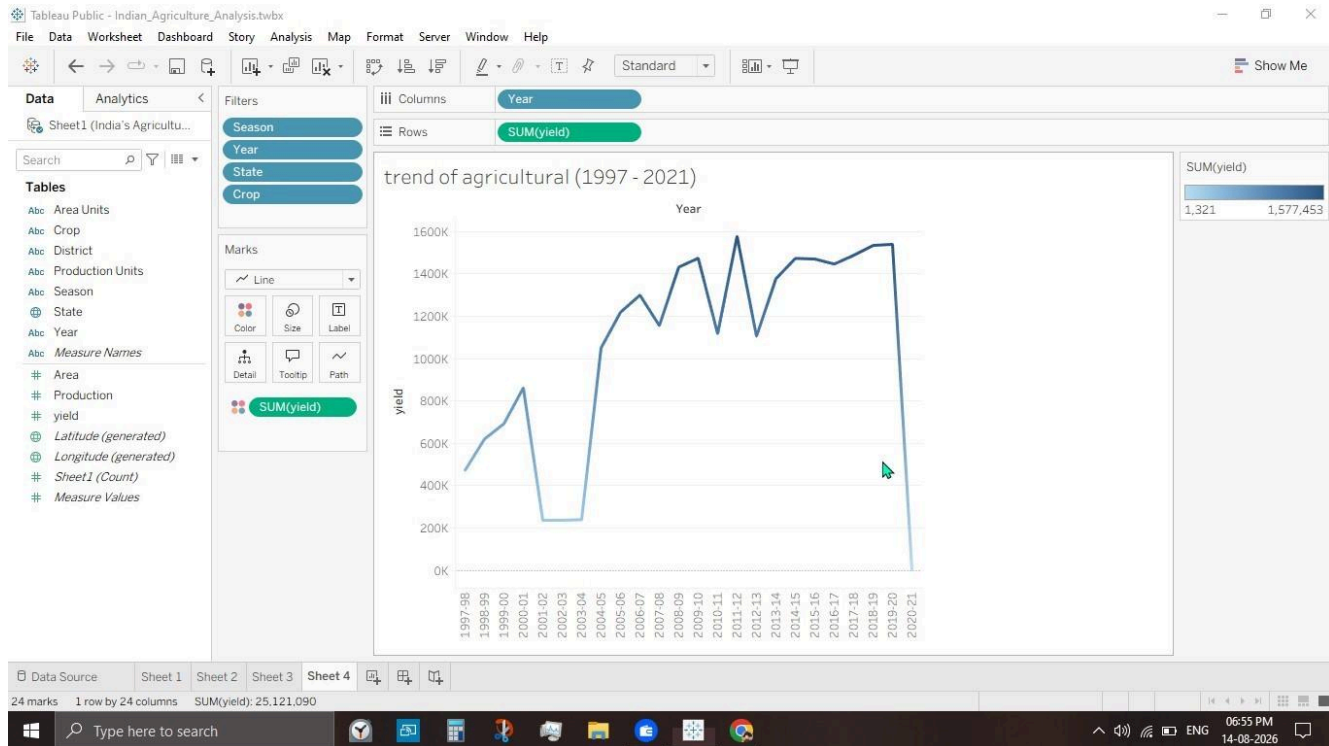
This helps compare production shares across seasons such as Kharif, Rabi and Autumn.

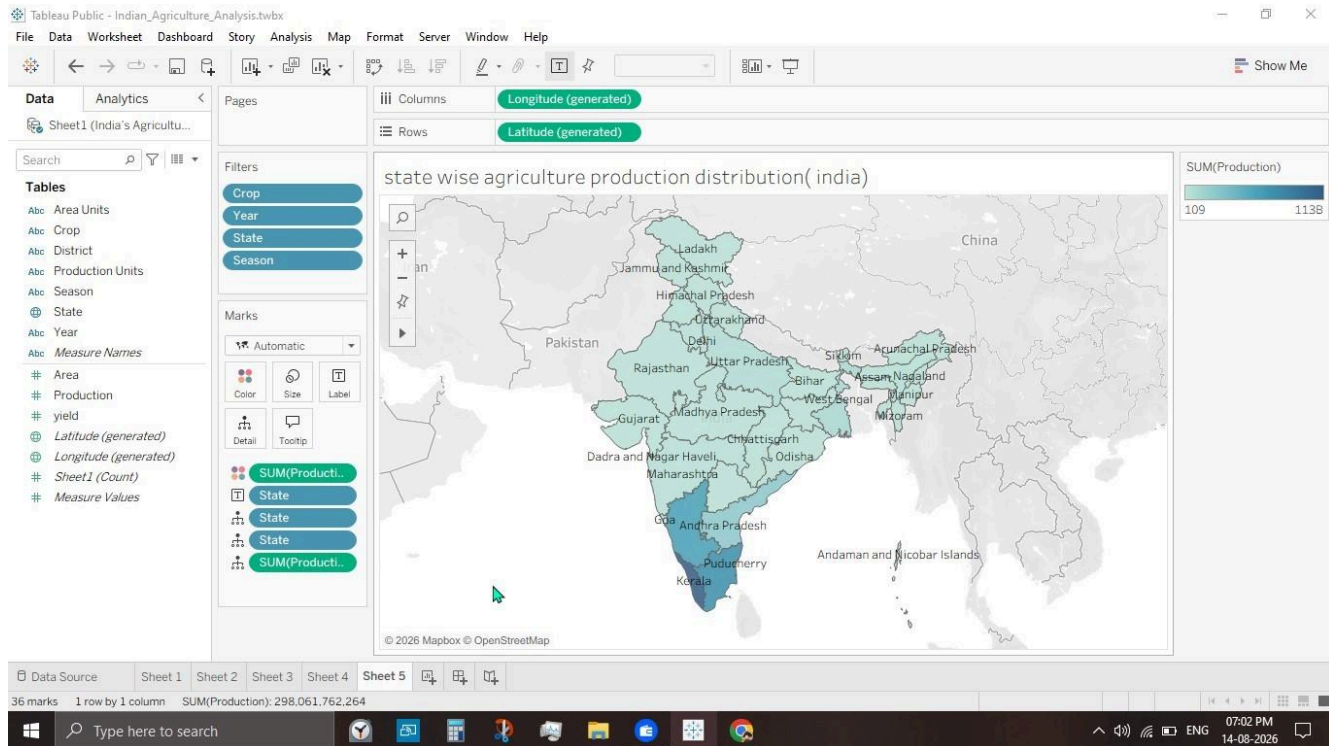
9. Which states and crops show better agricultural performance?

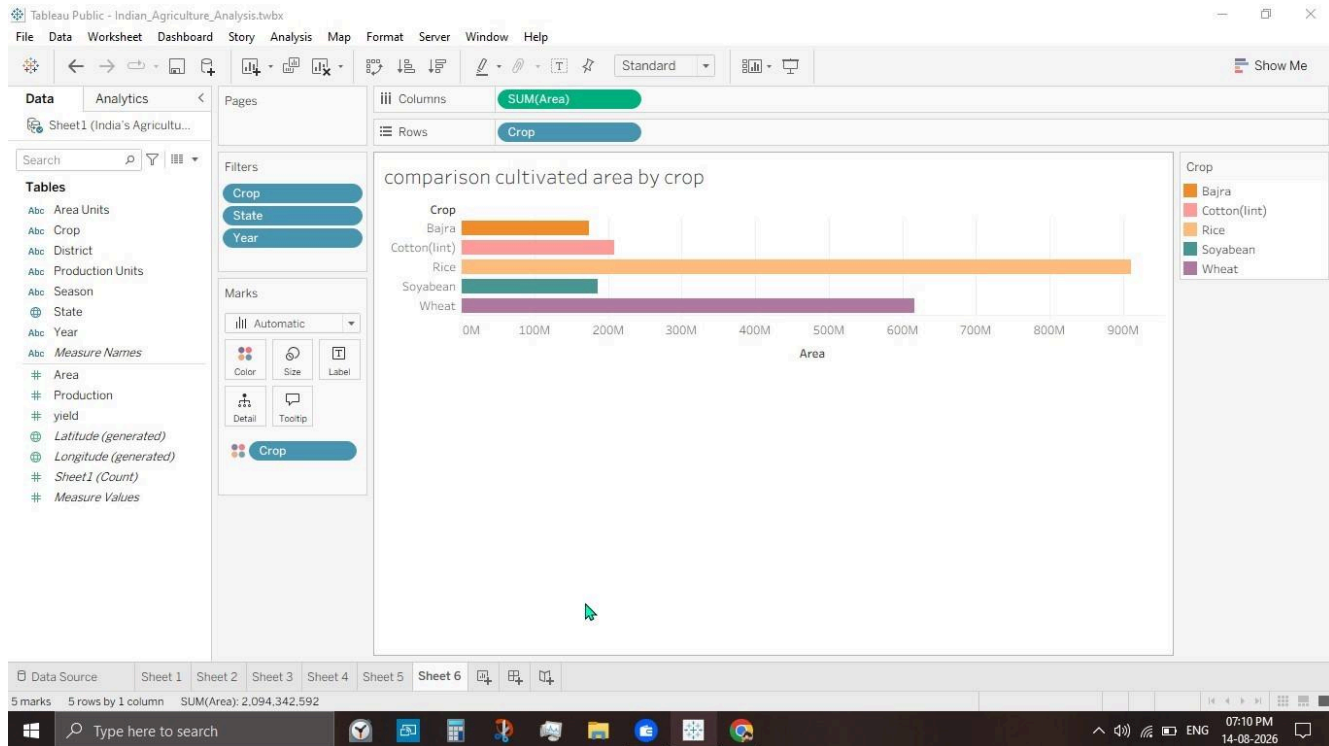
Description:

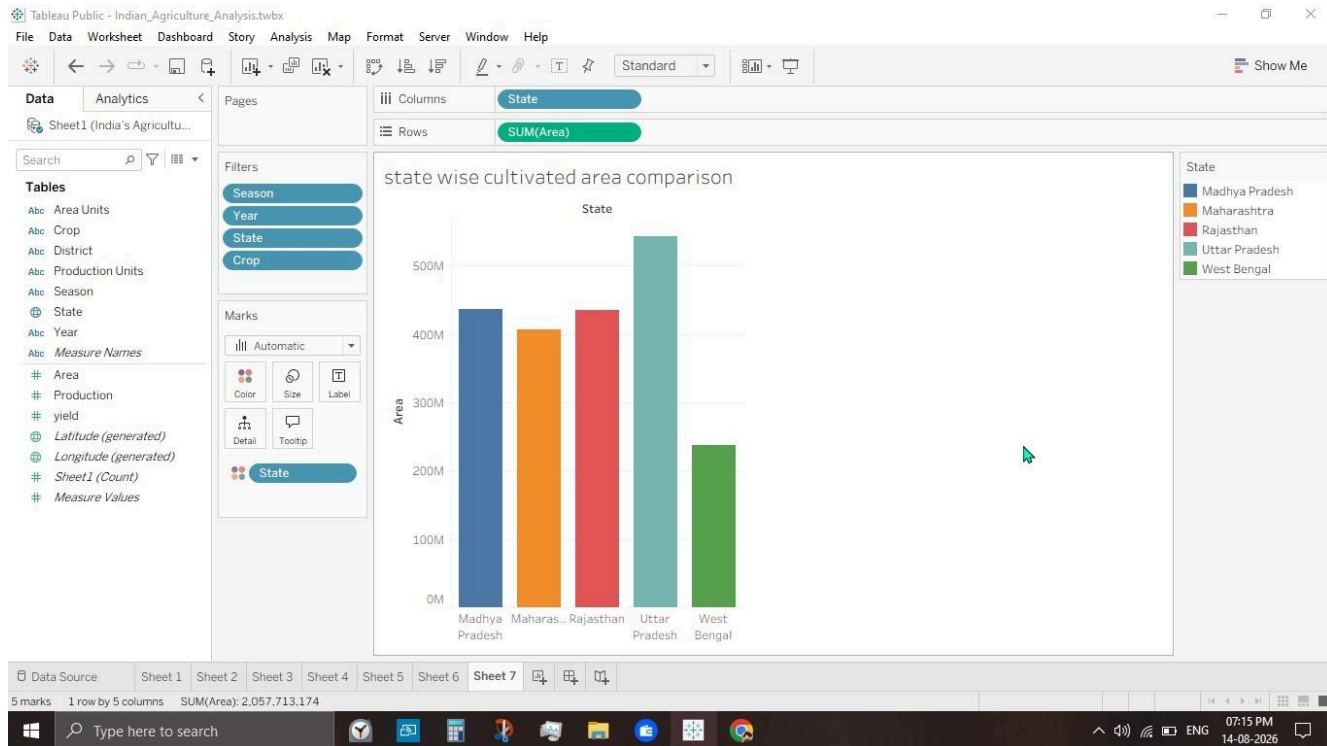
This question combines state and crop-level information to identify areas with stronger agricultural production and performance.

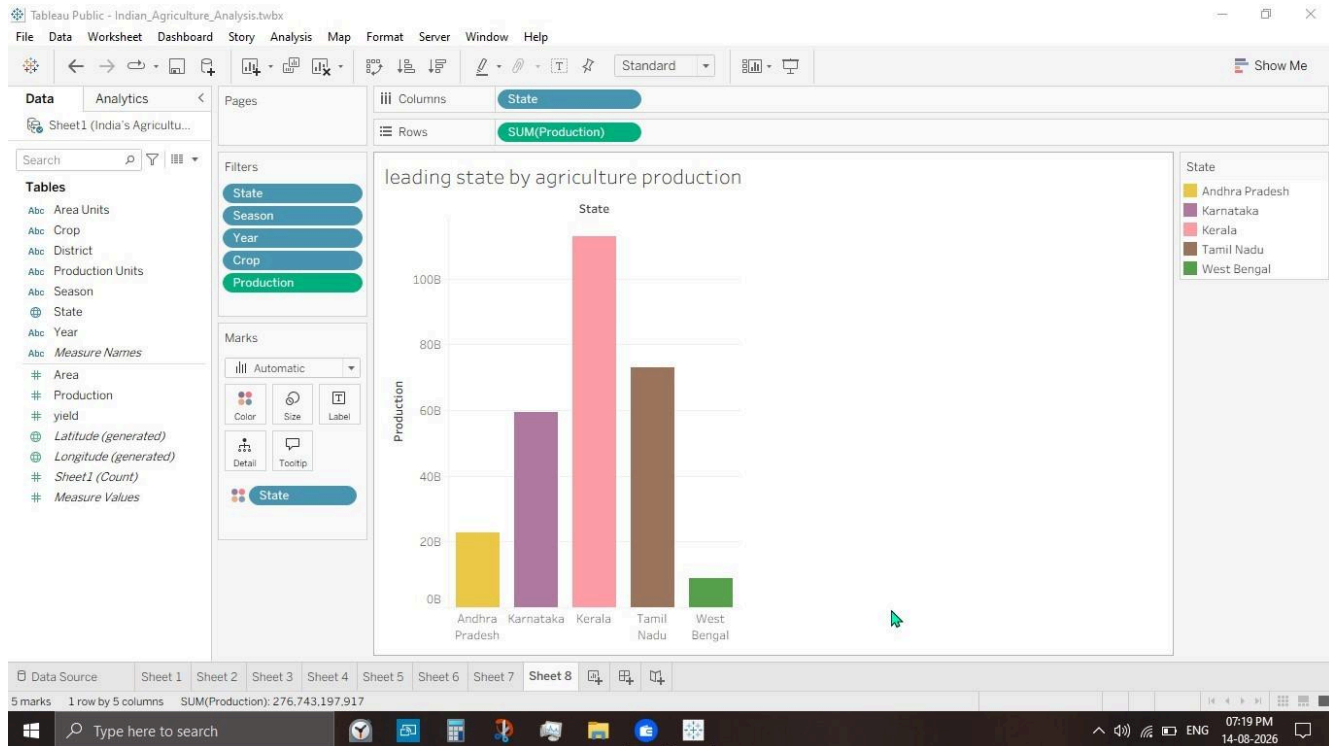
4.2. Developing Visualizations -

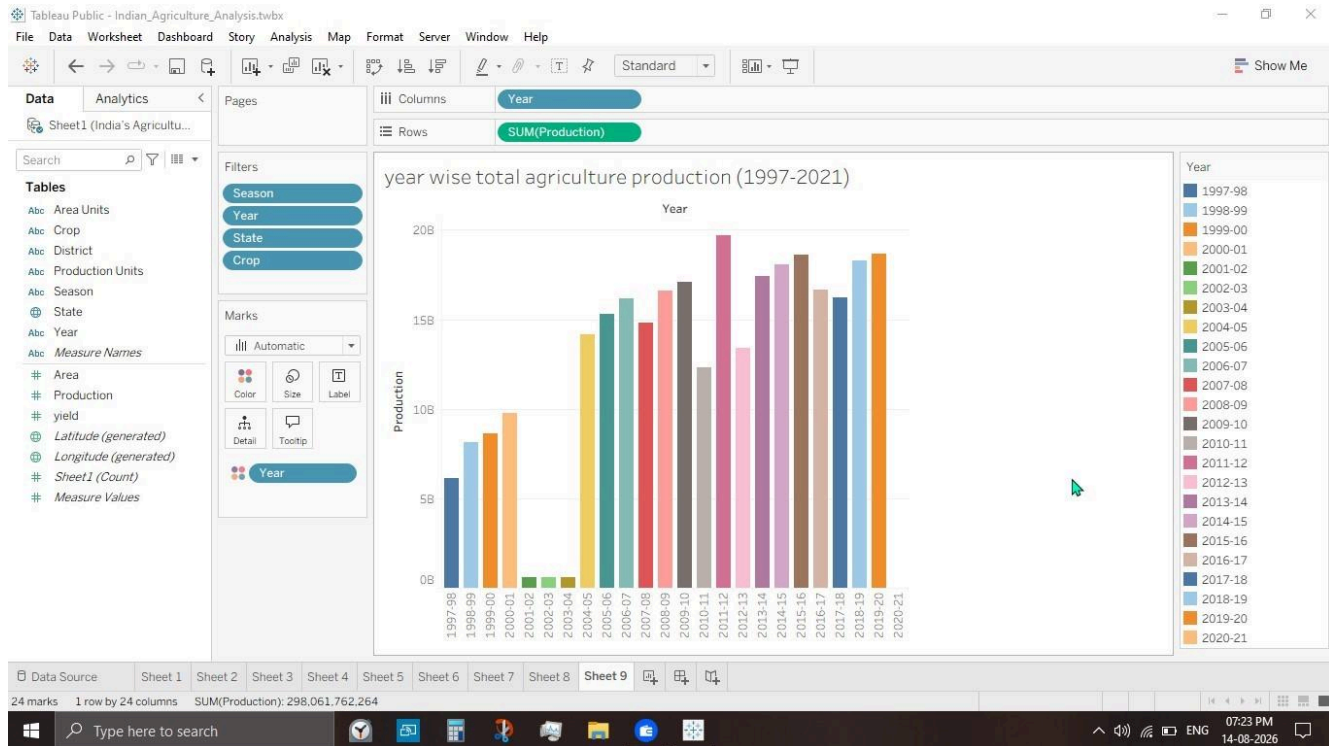


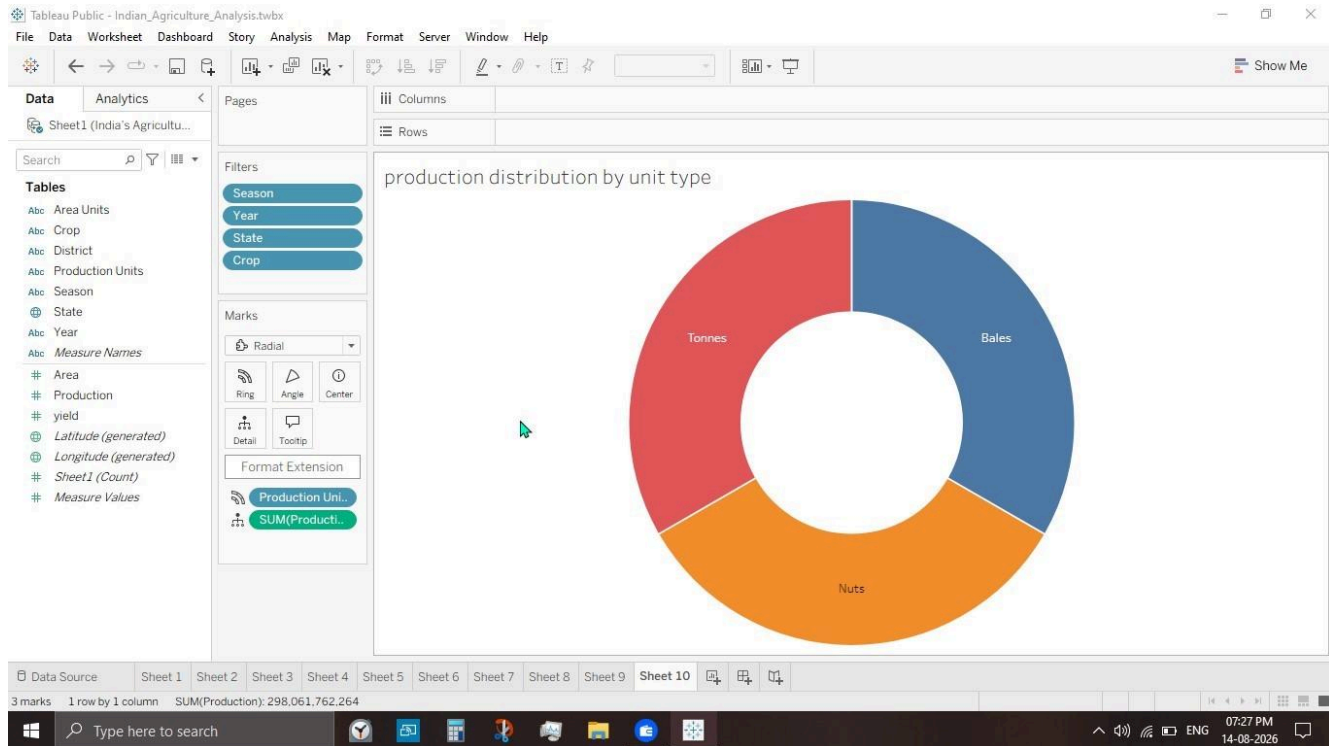


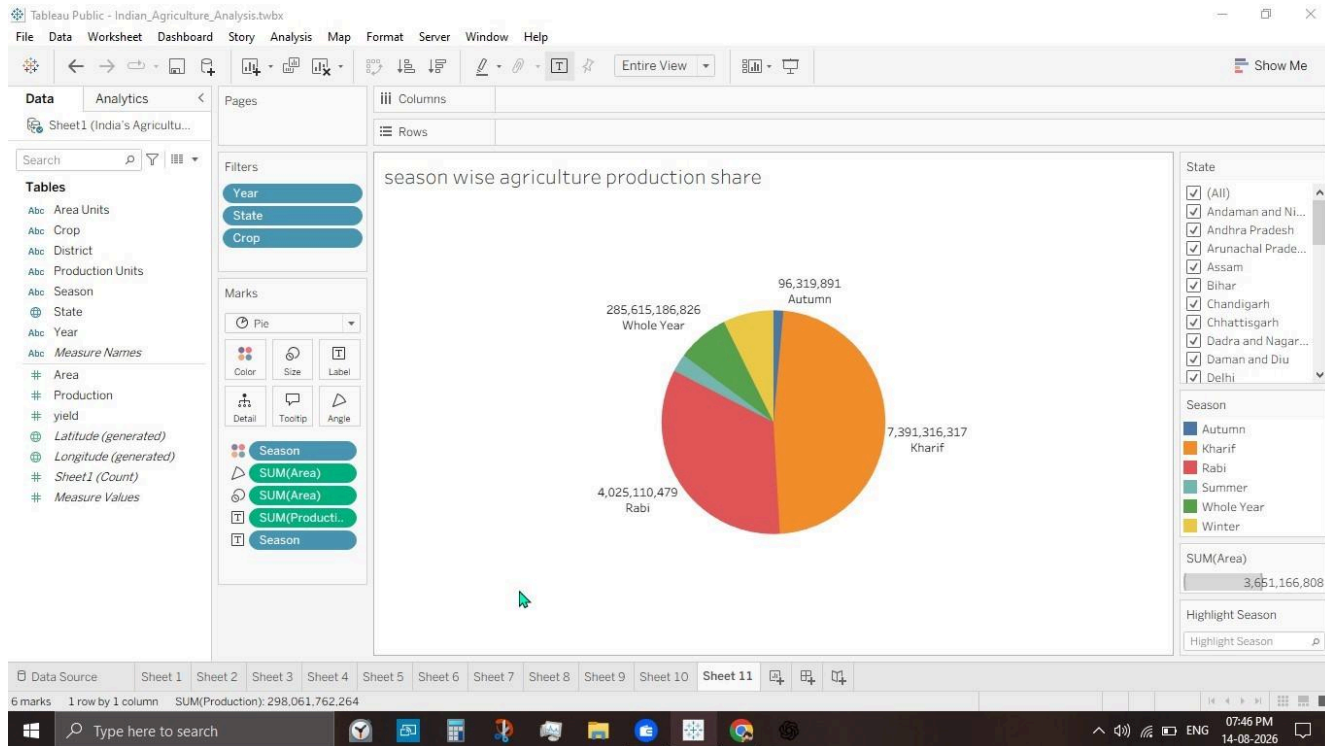










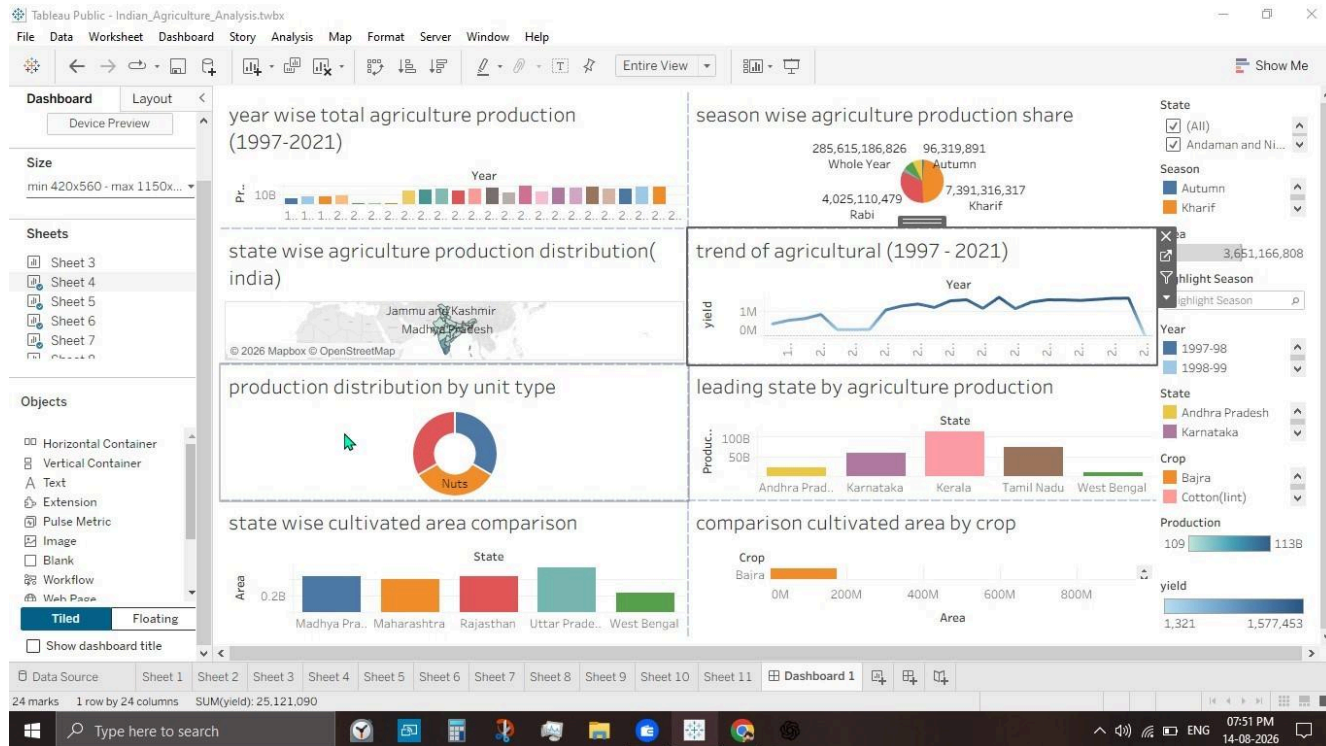


5. Dashboard

5.1. Dashboard Design File - 1. KPIs

- This section presents the key performance indicators of agricultural production. It helps users quickly understand the overall agricultural performance.
- 2. Trend of Agricultural Yield (1997–2021)
- This chart shows how agricultural yield changed from 1997 to 2021. It helps identify the overall trend and yearly variations in agricultural yield.
- 3. State-wise Agricultural Production Distribution
- This India map shows the distribution of agricultural production across different states. It helps identify regional differences in agricultural production.
- 4. Comparison of Cultivated Area by Crop

- This visualization compares the cultivated area of different crops. It helps identify which crops occupy a larger or smaller cultivated area.
- 5. State-wise Cultivated Area Comparison
- This chart compares cultivated areas across different states. It helps understand differences in agricultural land utilization among states.
- 6. Leading States by Agricultural Production
- This chart compares selected leading states based on agricultural production. It helps identify states with higher production levels.
- 7. Year-wise Total Agricultural Production (1997–2021)
- This chart presents total agricultural production year by year from 1997 to 2021. It helps understand changes in overall production over time.
- 8. Production Distribution by Unit Type
- This visualization shows how agricultural production is distributed according to the unit types available in the dataset. It helps understand the production measurement distribution.



6. Report

6.1. Story Design File - Observations

Agricultural production varies across different years, showing changes in India's overall agricultural performance.

Agricultural production is not equally distributed among all states. Some states contribute significantly more to total production than others.

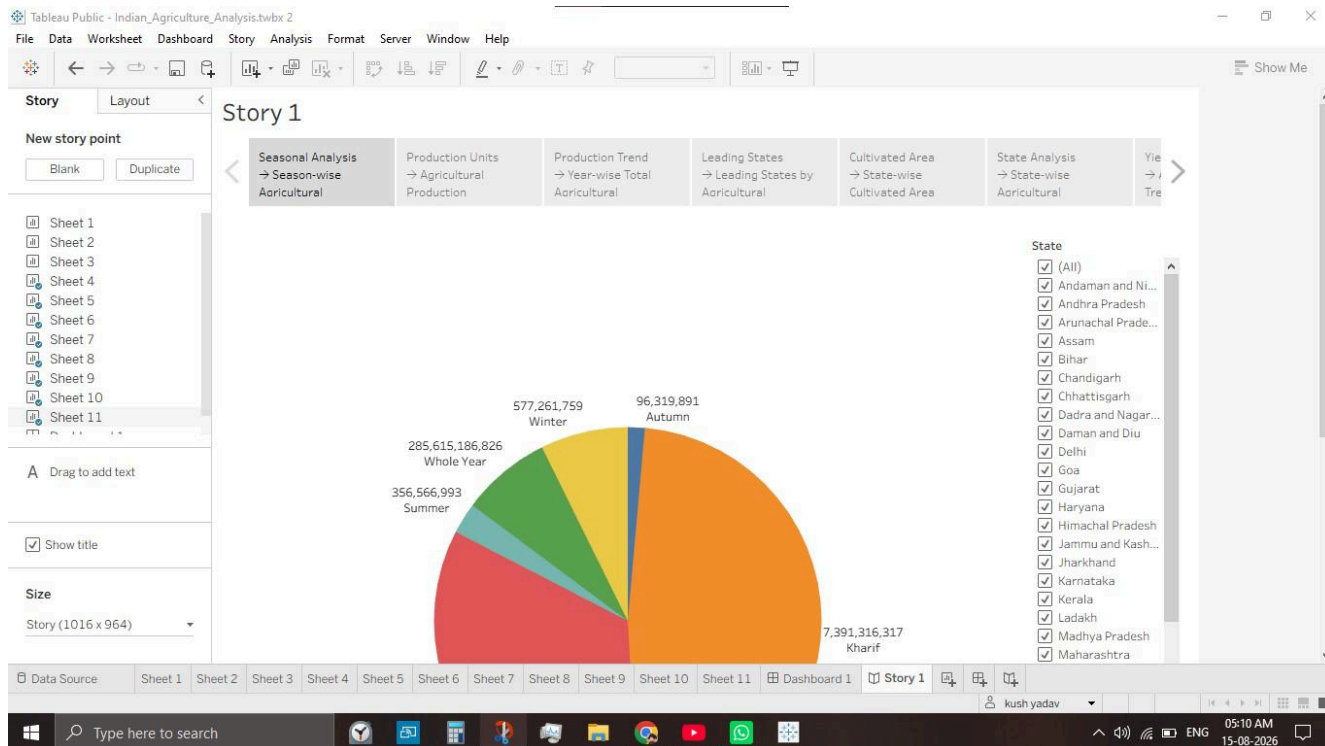
Cultivated area differs across crops, indicating that some crops are grown over larger agricultural areas.

States also show differences in cultivated area, reflecting variation in agricultural land utilization.

Season-wise production is different, with Kharif, Rabi and Autumn contributing different shares to agricultural production.

The dashboard shows clear differences in crop, state and year-wise performance, making it easier to compare agricultural production.

Overall, the visualization helps identify important production patterns and trends in India's agricultural sector.



7. Performance Testing

7.1 Utilization of Data Filters

Data filters are used to allow users to view agricultural information according to different categories. Users can explore the data based on available fields such as year, state, crop and season.

7.2 No. of Calculation Fields

Calculated fields were used where required to support the analysis and visualization. They helped in creating meaningful measures and presenting useful agricultural insights.

7.3 No. of Visualization

The dashboard contains multiple visualizations, including KPI cards, line charts, bar charts, maps and comparison charts. These visualizations present agricultural production from different perspectives.

8. Conclusion/Observation - The project provides a clear analysis of India's agricultural crop production using Tableau. The dashboard shows differences in production across states, crops,

years, cultivated areas and seasons. The visualizations make complex agricultural data easier to understand and help identify important trends and patterns.

9. Future Scope - The project can be improved by adding more recent agricultural data and additional crop-related information. Future versions can include forecasting, advanced filters and more detailed state-level analysis. Machine learning techniques can also be used to predict future agricultural production and identify possible trends.

10. Appendix -

10. GitHub & Project Demo Link – github link -

<https://github.com/kush-yadav7424/indian-agriculture-crop-production-analysis-using-by-tableau>

Demo link

https://drive.google.com/file/d/1Wqwg-0WnzlL0mVmeh3LKsk-14_DuS9Sj/view?usp=drive_link